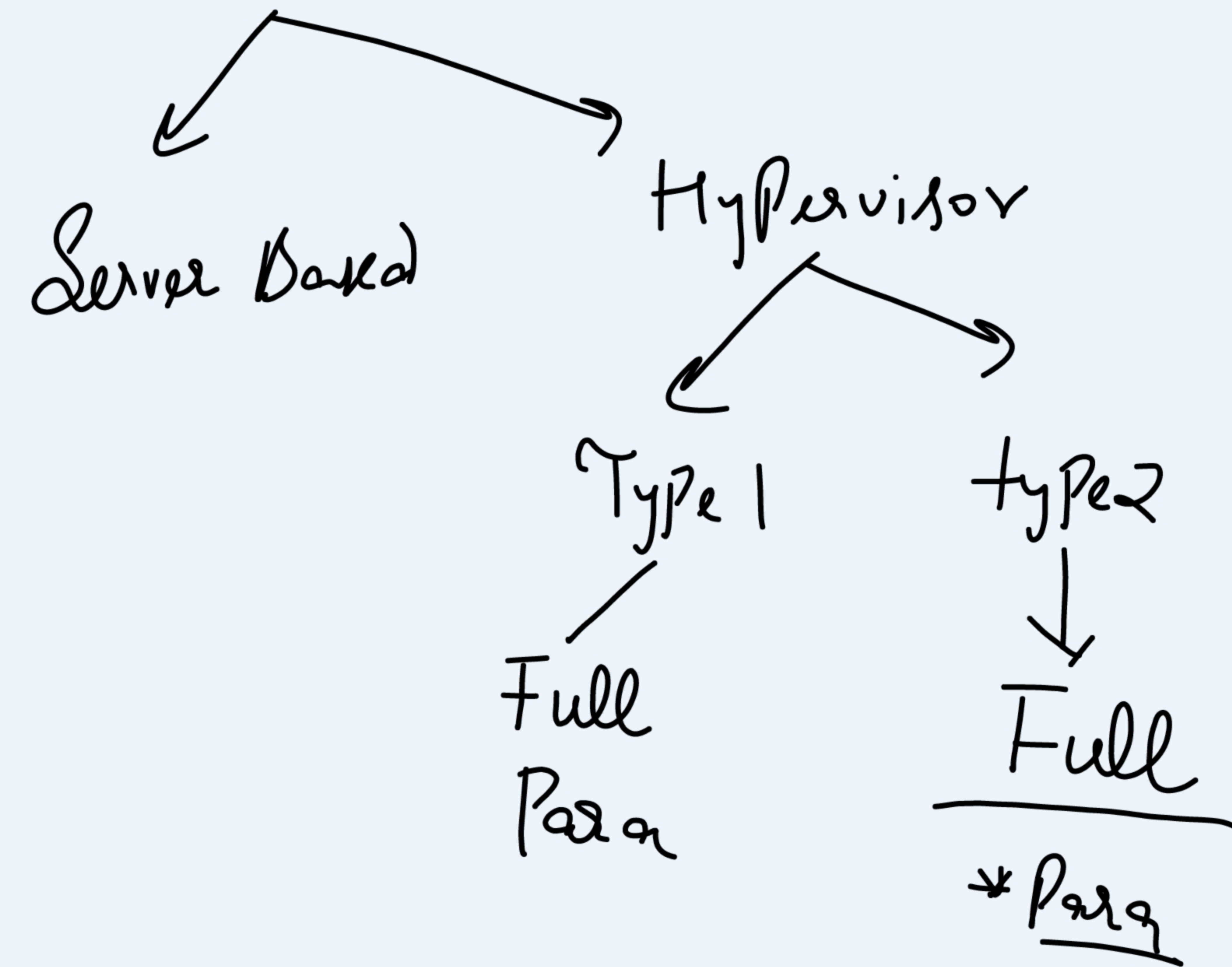
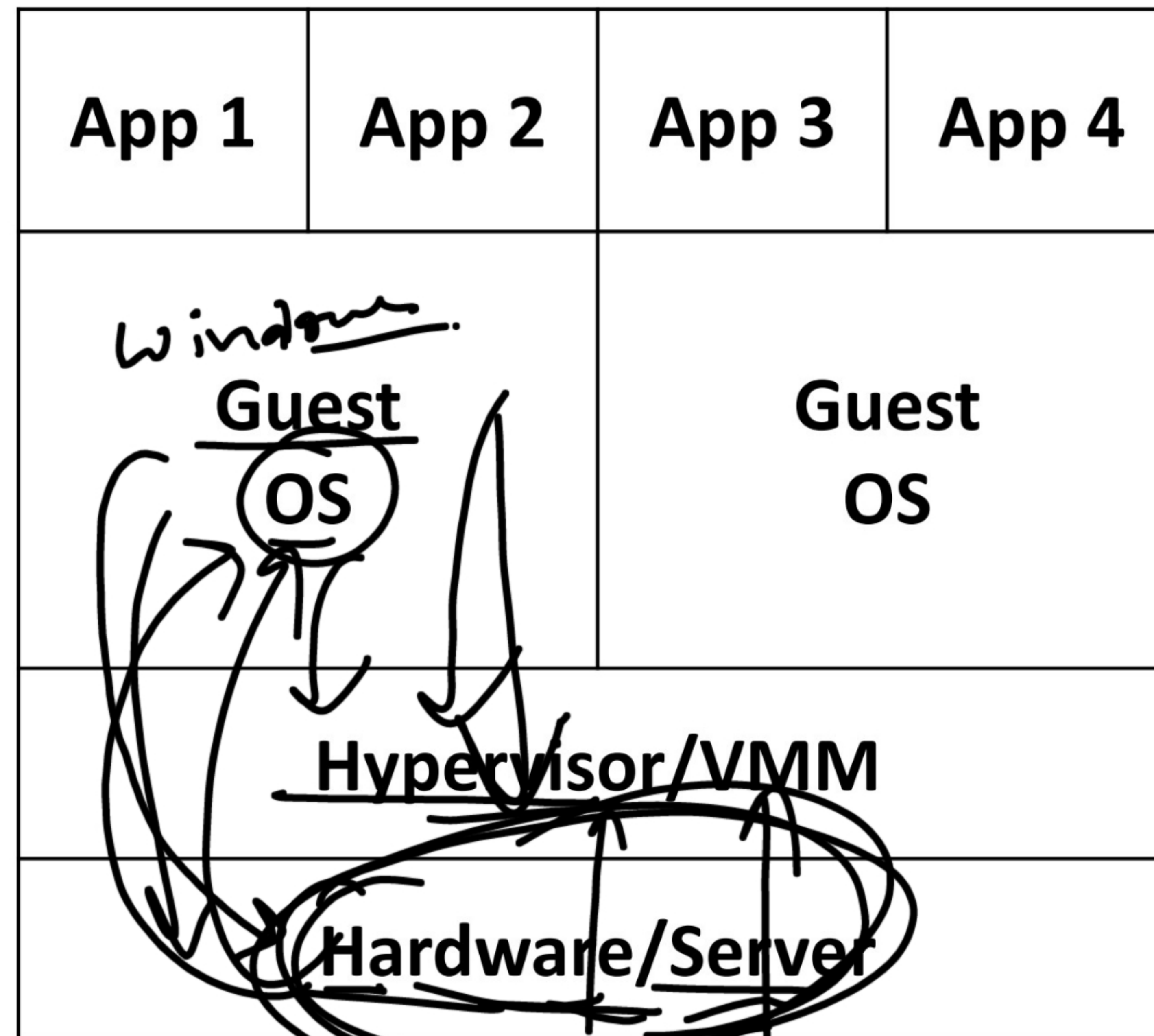


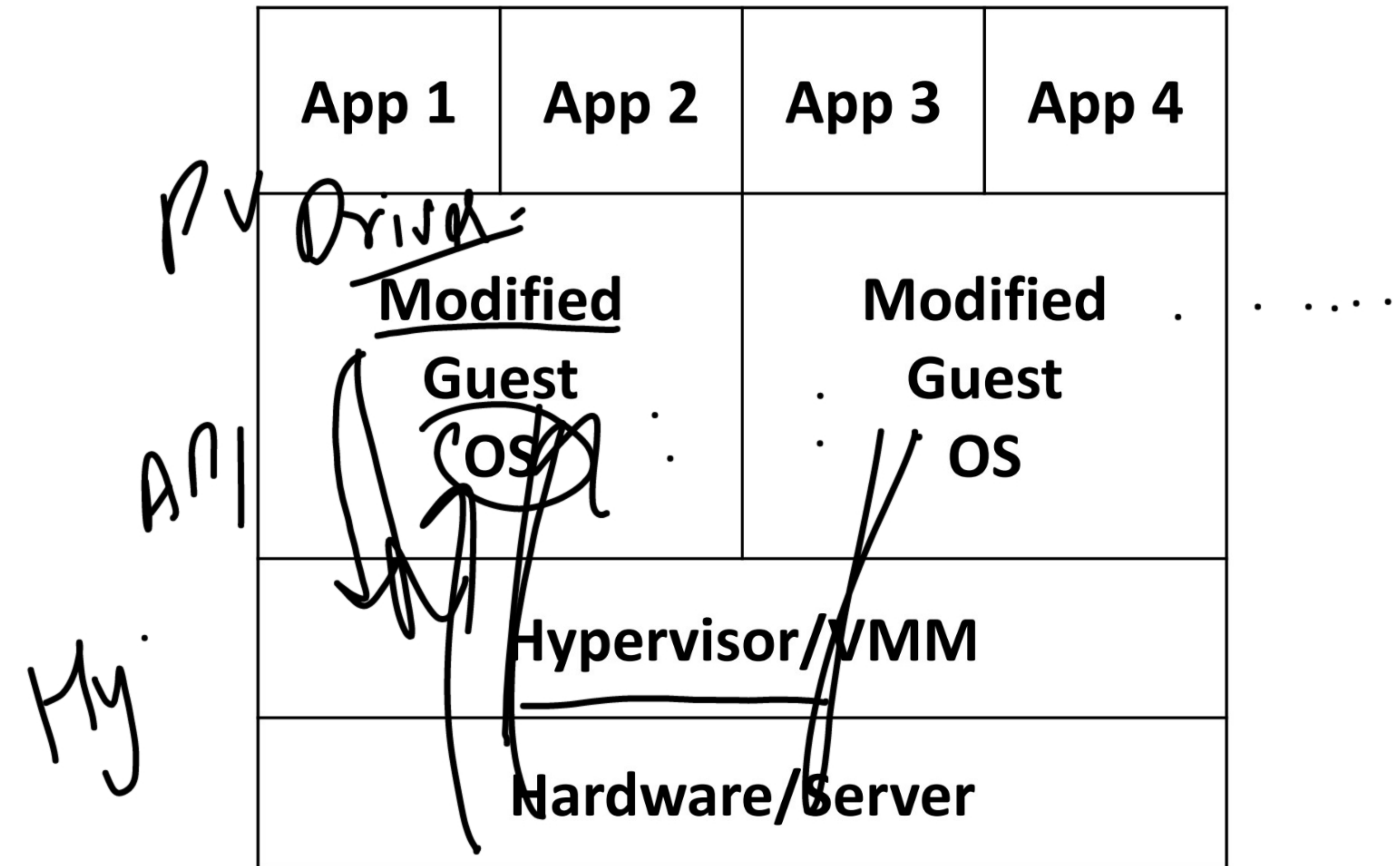
Full vs Para Virtualization

- In full virtualization, the guest operating system (OS) is not aware that it's running in a virtualized environment.
- The hypervisor emulates the hardware entirely, so the guest OS interacts as if it's on actual hardware.
- In paravirtualization, the guest OS is aware that it's running in a virtualized environment and is modified to work with the hypervisor directly.
- This allows more efficient communication between the guest OS and the hypervisor without full hardware emulation.





Full Virtualization



Para Virtualization

Request and Response Mechanisms

- **Full Virtualization:**

- Relies on hardware emulation.
- Hypervisor intercepts and translates requests and responses.
- Increased overhead and slower response due to the emulation layer.

- **Paravirtualization:**

- Uses direct Hypercalls between the guest OS and the hypervisor.
- Bypasses the emulation layer, allowing faster request and response handling.
- Lower overhead and better performance for supported OSes.

- Full Virtualization is often preferred in environments where compatibility and ease of setup are important (e.g., running various OS types in a lab or for application testing). It's also ideal for users who need strong isolation between VMs without modifying various guest OS.
- Paravirtualization is preferred in data centers and high-performance environments where compatible OSes are common (such as Linux servers) and where maximizing performance and reducing overhead are crucial.